

MODULE 23

SQL Backup Recovery and Maintenance

DBMS with SQL - Enterprise Learning Series

ARSHITH LMS | Professional Edition

SQL Backup Recovery and Maintenance

Backup Types

Full: complete copy of everything

Incremental: only changes since last backup

Differential: changes since last FULL backup

Logical: SQL dump file (portable)

Physical: raw database files

mysqldump

-- Backup single database

```
mysqldump -u root -p school_db > backup.sql
```

-- Backup all databases

```
mysqldump -u root -p --all-databases > all.sql
```

-- Restore from backup

```
mysql -u root -p school_db < backup.sql
```

-- Compressed backup

```
mysqldump -u root -p school_db | gzip > backup.sql.gz
```

Maintenance Commands

CHECK TABLE students; -- check for errors

REPAIR TABLE students; -- repair corrupted table

OPTIMIZE TABLE students; -- reclaim space

ANALYZE TABLE students; -- update key stats

SHOW TABLE STATUS LIKE 'students';

Performance Monitoring

SHOW PROCESSLIST;

SHOW STATUS LIKE 'Connections';

SHOW VARIABLES LIKE 'slow_query_log%';

SET GLOBAL slow_query_log = 'ON';

SET GLOBAL long_query_time = 2;

Replication Basics

Master: handles all writes

Slave: handles reads (improves performance)

Benefits: high availability, load balancing,
backups without downtime, geographic distribution

Types:

Asynchronous: master does not wait

Synchronous: master waits for confirmation